CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 11: **Exploring Some Geometric Themes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A polyhedron has 6 triangular faces and 3 quadrilateral faces and no other faces. Its number of vertices is: 1

(a) 26 **(b)** 8 **(c)** 15 **(d)** 9

2. All the 24 faces of a polyhedron are pentagons. Its number of vertices is: 1

(a) 38 **(b)** 60 **(c)** 86 **(d)** 36

3. A polyhedron has 12 edges and exactly 4 edges meet at every vertex. Its number of faces is: 1

(a) 8 **(b)** 10 **(c)** 14 **(d)** 6

4. If one corner of a cube is sliced off, the number of vertices of the new solid is: 1

(a) 9 **(b)** 8 **(c)** 11 **(d)** 10

5. A cube is painted all over and cut into unit cubes. If exactly 216 of the unit cubes have exactly one face painted, the side of the big cube is: 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 7

6. A Koch curve starts with a 9 cm line. The length of the curve at stage 2 (in cm) is: 1

(a) 13 **(b)** 16 **(c)** 25 **(d)** 18

7. A Sierpinski triangle starts with a triangle of side 160 cm. At stage 4 the side of each small shaded triangle is (in cm): 1

(a) 40 **(b)** 5 **(c)** 20 **(d)** 10

8. A polyhedron has 2 triangular faces and 3 quadrilateral faces and no other faces. Its number of vertices is: 1

(a) 5 **(b)** 9 **(c)** 6 **(d)** 16

9. **Assertion (A):** A polyhedron with 10 vertices and 15 edges has 7 faces. 1
Reason (R): For a polyhedron, $F = 2 - V + E$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** When a $3 \times 3 \times 3$ cube painted on all faces is cut into 27 unit cubes, exactly one small cube has no painted face. 1
Reason (R): Eight of the 27 small cubes have exactly three painted faces.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A Sierpinski triangle starts with a triangle of side 160 cm. Find the number of shaded triangles and the side of each at stage 4. 2

12. A cube is painted all over and cut into unit cubes. If 120 unit cubes have exactly two painted faces, find the side of the cube and the total number of unit cubes. 2

13. A polyhedron has 6 triangular faces and 8 square faces only. Find its number of edges. 2

14. A Sierpinski triangle starts with a triangle of side 144 cm. Find the number of shaded triangles and the side of each at stage 4. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Every face of a polyhedron is a triangle and it has 48 faces. Find the number of edges and vertices and verify Euler's formula. 3

16. A cube is painted all over and cut into unit cubes. For which side is the number of unit cubes with exactly two painted faces equal to 264? Hence find the number with exactly one painted face. 3

17. In the Sierpinski triangle construction, find the first stage at which the number of shaded triangles exceeds 20000. How many shaded triangles are there at that stage? 3

OR

Every face of a polyhedron is a triangle and it has 52 faces. Find the number of edges and vertices and verify Euler's formula. 3

18. A cube is painted all over and cut into unit cubes. For which side is the number of unit cubes with exactly two painted faces equal to 336? Hence find the number with exactly one painted face. 3

OR

In the Sierpinski triangle construction, find the first stage at which the number of shaded triangles exceeds 100000. How many shaded triangles are there at that stage? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher draws a Sierpinski triangle. Start with a solid triangle; at stage 1 the middle triangle is removed leaving 3 triangles. At each next stage the middle part of every remaining triangle is removed in the same way. 5

(i) How many triangles remain at stage 1?

(ii) How many remain at stage 2?

(iii) How many remain at stage 3?

(iv) Write the number at stage n and find it at stage 5.

20. A polyhedron has 6 triangular faces and 12 quadrilateral faces and no other faces. Find the number of edges and vertices, and verify Euler's formula. 5

OR

A Koch snowflake is built from an equilateral triangle of side 36 cm. Find its perimeter at stage 2, and by how much the perimeter has increased from stage 0. 5

END OF QUESTION PAPER — Total: 40 Marks